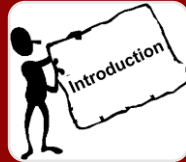


On your tutorials



UNIVERSIDAD
DE MÁLAGA





What is flamapy?



Flamapy as a tool



flamapy as a framework

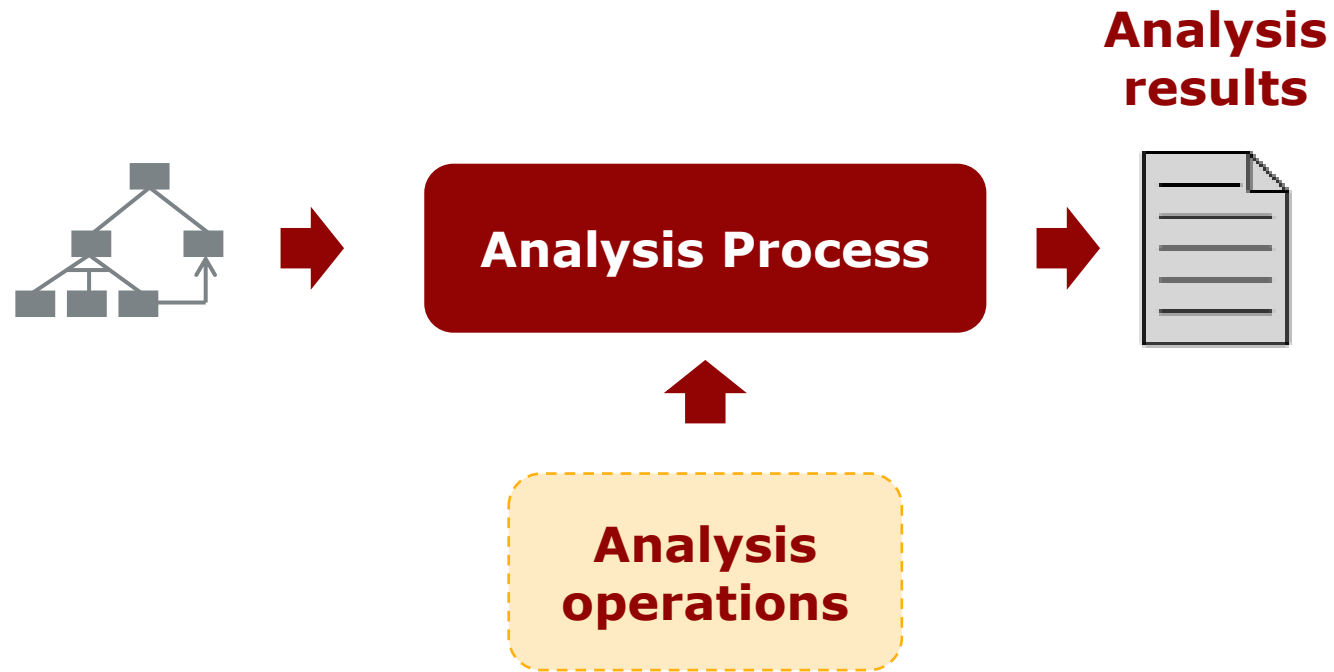


Let's get dirty

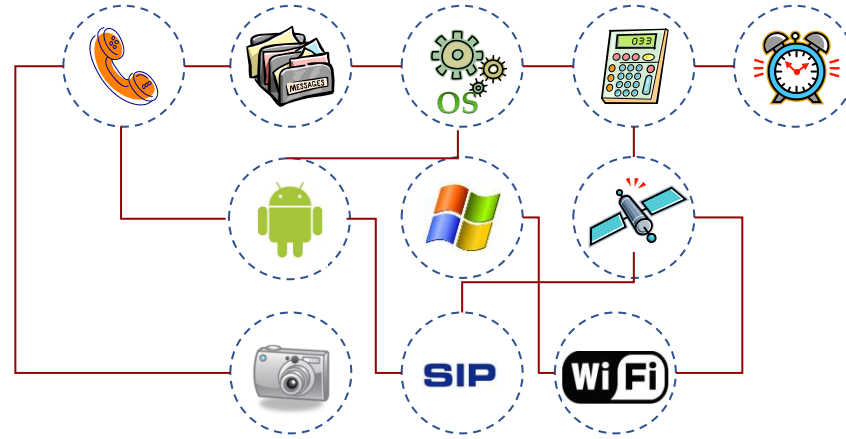
Variability Models



Automated Analysis



Analysis Operations



**My app requires
wifi and bluetooth 4**



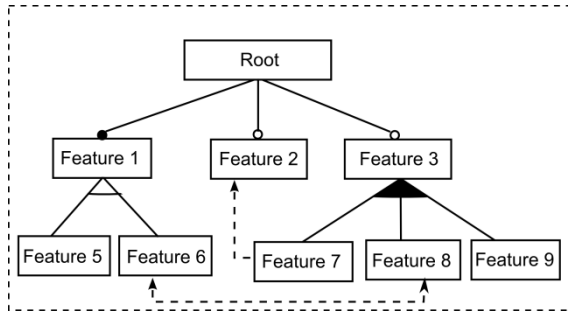
**In how many different
devices will it work?**



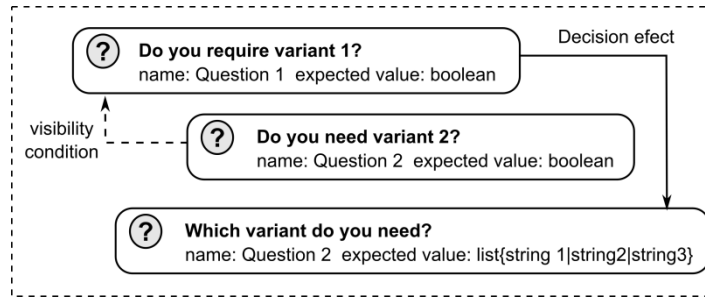
A tool and a framework for the

Automated Analisis Of Variability Models

Variability models... but feature modelling



Feature modelling

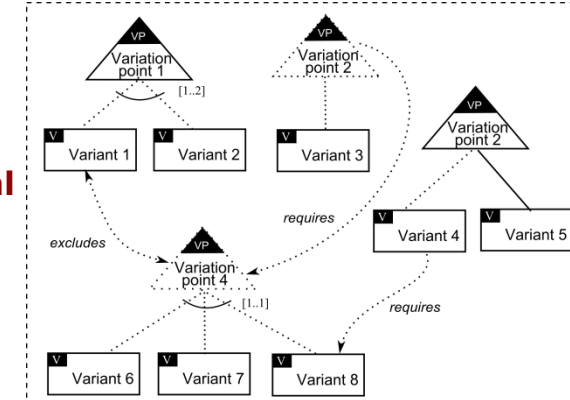


Decision modelling

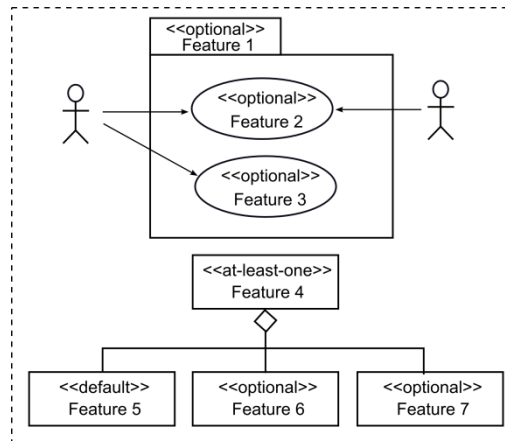
Ad-hoc solutions:
tables, textual
docs, ...

Techniques

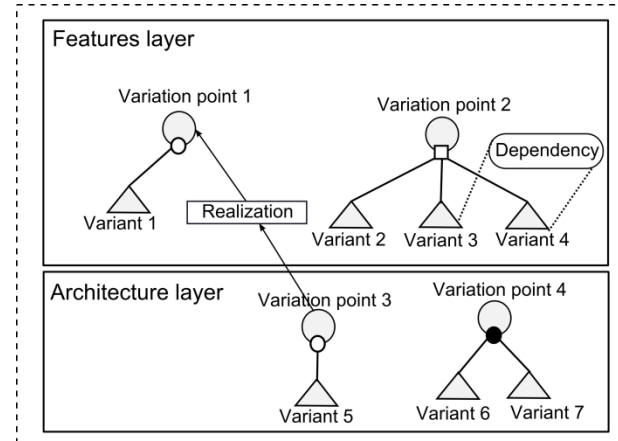
Orthogonal
variability
modelling



UML-based



COVAMOF



A tool and a framework

As a tool

Allows to analyze feature models

Multiple interfaces

- CMD
- Python
- WASM

As a framework

Adapt it to what you need

- Allow other variability models
- Create your:
 - Operations
 - Transformations
 - Solvers

Integrate it in your tools

Flama as a tool

- Simple installation
 - `pip install flamapy`

Supported formats

- UVL
- Fama XML
- Feature IDE
- Glencoe
- Others (CNF, JSON, ...)

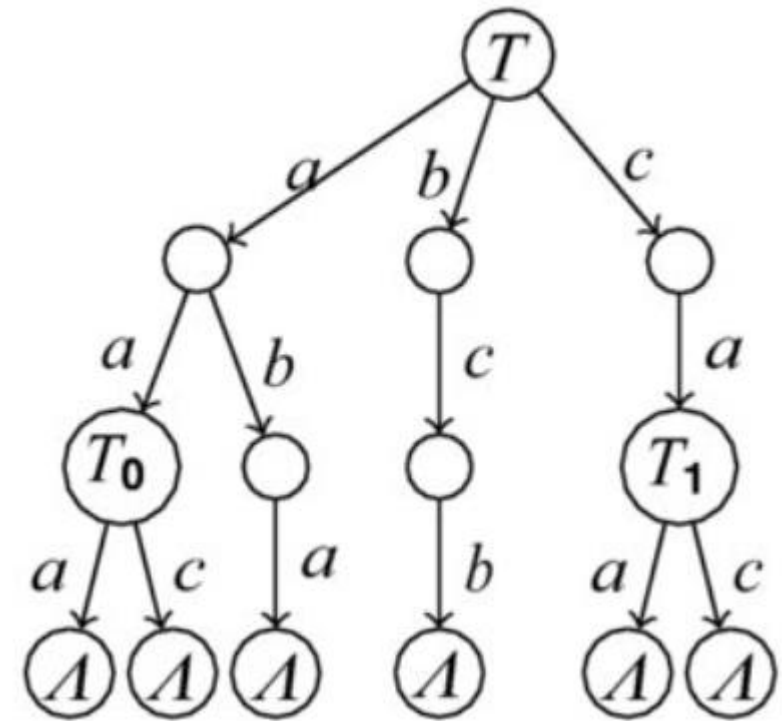
Analysis operations

Atomic Sets	Average Branching Factor	Commonality	Configurations	Number of Configurations
Conflict Detection	Core Features	Count Leafs	Dead Features	Estimated Number of Configurations
False Optional Features	Feature Ancestors	Filter	Leaf Features	Max Depth
	Satisfiable	Satisfiable Configuration	Unique Features	

Solvers supported

PySAT 

CaDiCaL (rel-1.0.3)
CaDiCaL (rel-1.5.3)
CaDiCaL (rel-1.9.5)
Glucose (3.0)
Glucose (4.1)
Glucose (4.2.1)
Lingeling (bbc-9230380-160707)
MapleLCMDistChronoBT (SAT competition 2018 version)
MapleCM (SAT competition 2018 version)
Maplesat (MapleCOMSPS_LRB)
Mergesat (3.0)
Minicard (1.2)
Minisat (2.2 release)
Minisat (GitHub version)

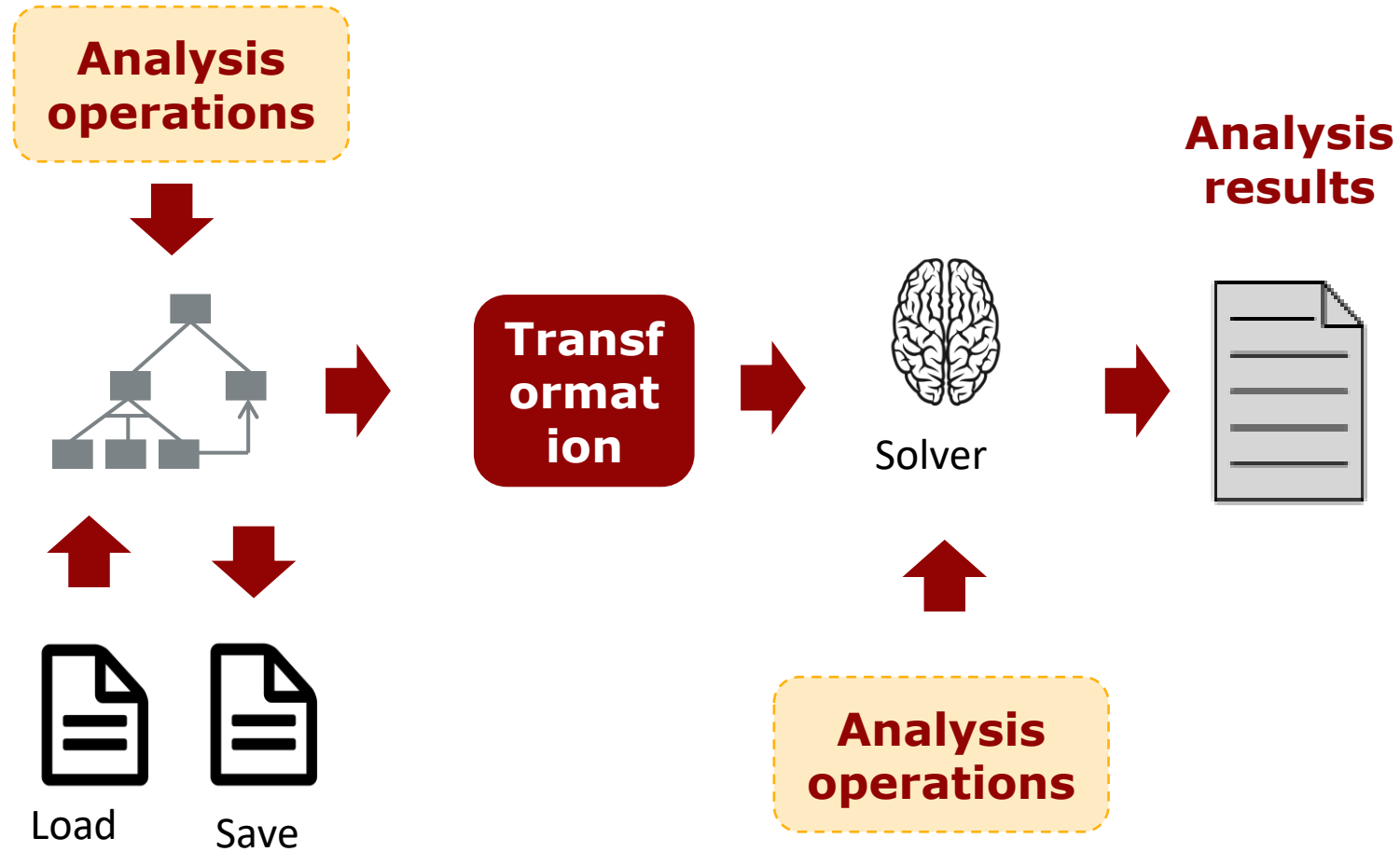


CUDD Solver

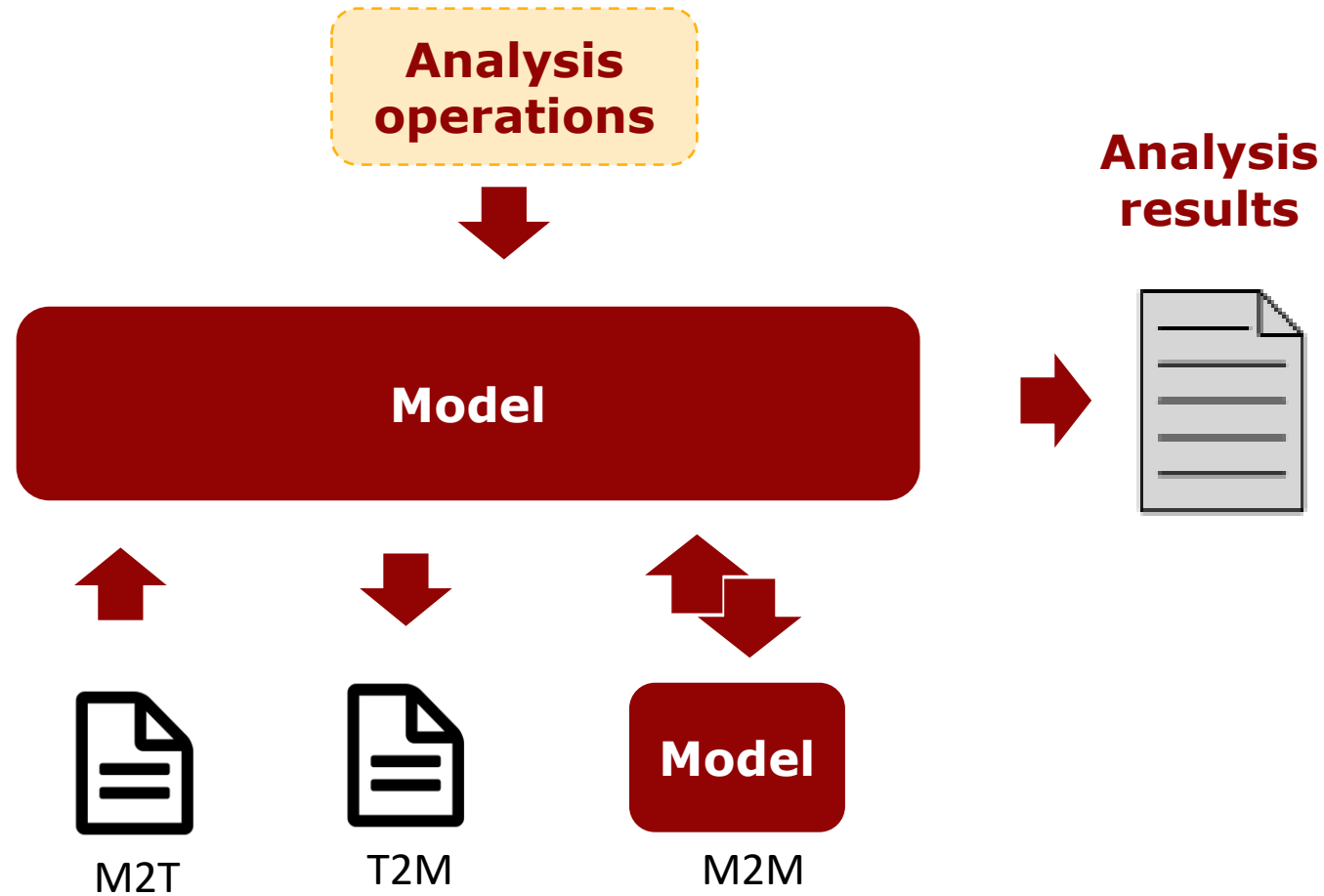
Interfaces

- Use it easily and fast
 - <https://www.flamapy.org/try-it/wasm-uvi-analysis>

The old metamodeling approach



The “new” metamodeling approach



FM metamodel

Count leafs
Count products
approximated
...

FM metamodel

**Analysis
results**

- Fide writer
- UVL writer
- ...
- UVL reader
- FamaXML reader
- SPLX reader
- ...

**m2
m**

• ...

Pysat metamodel

#Products
Valid configuration
Valid Product
...

**PySAT
metamodel**

**Analysis
results**

- CNF writer
- ...
- CNF reader
- ...
- Feature model to SAT

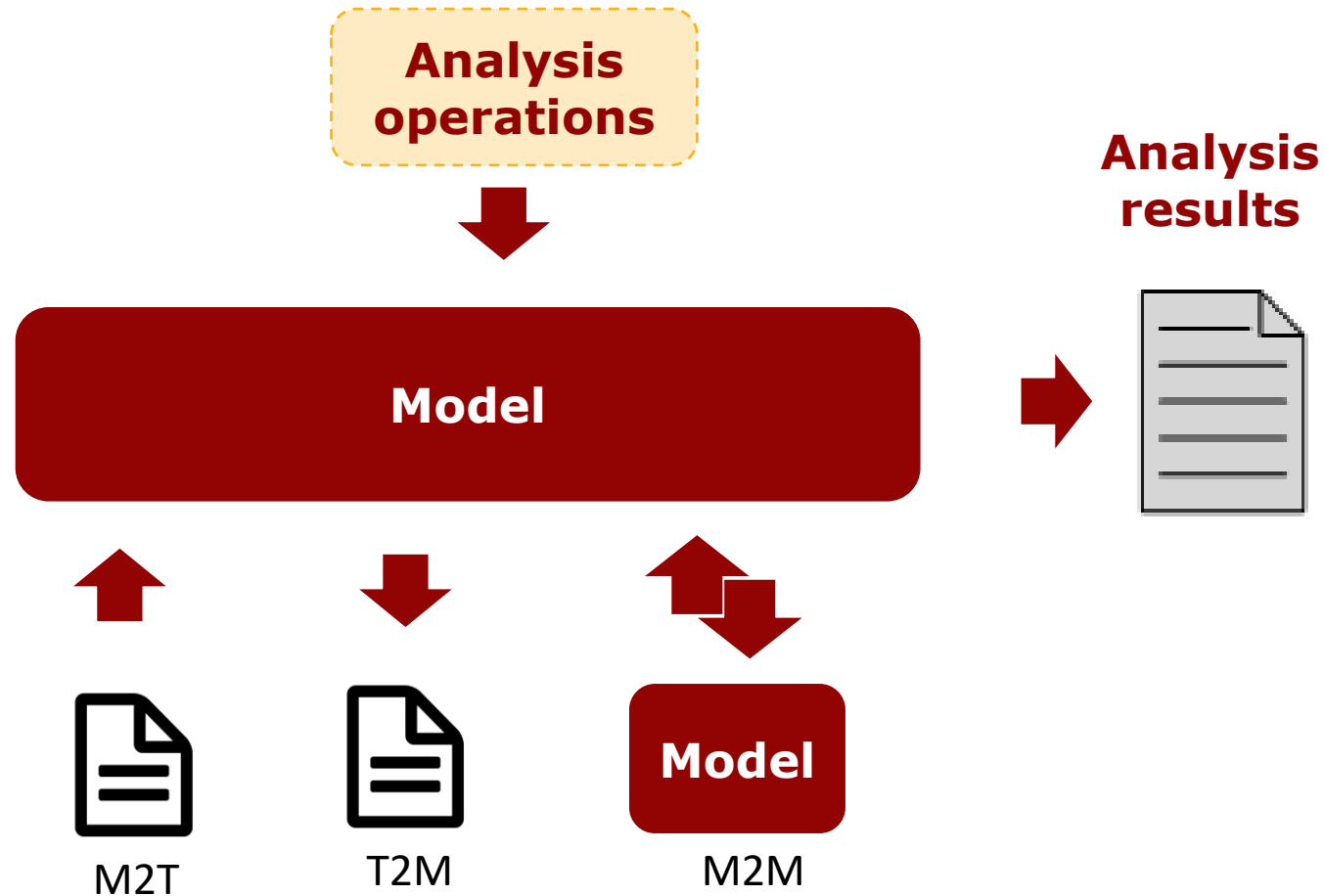
**m2
m**

• Feature model to SAT

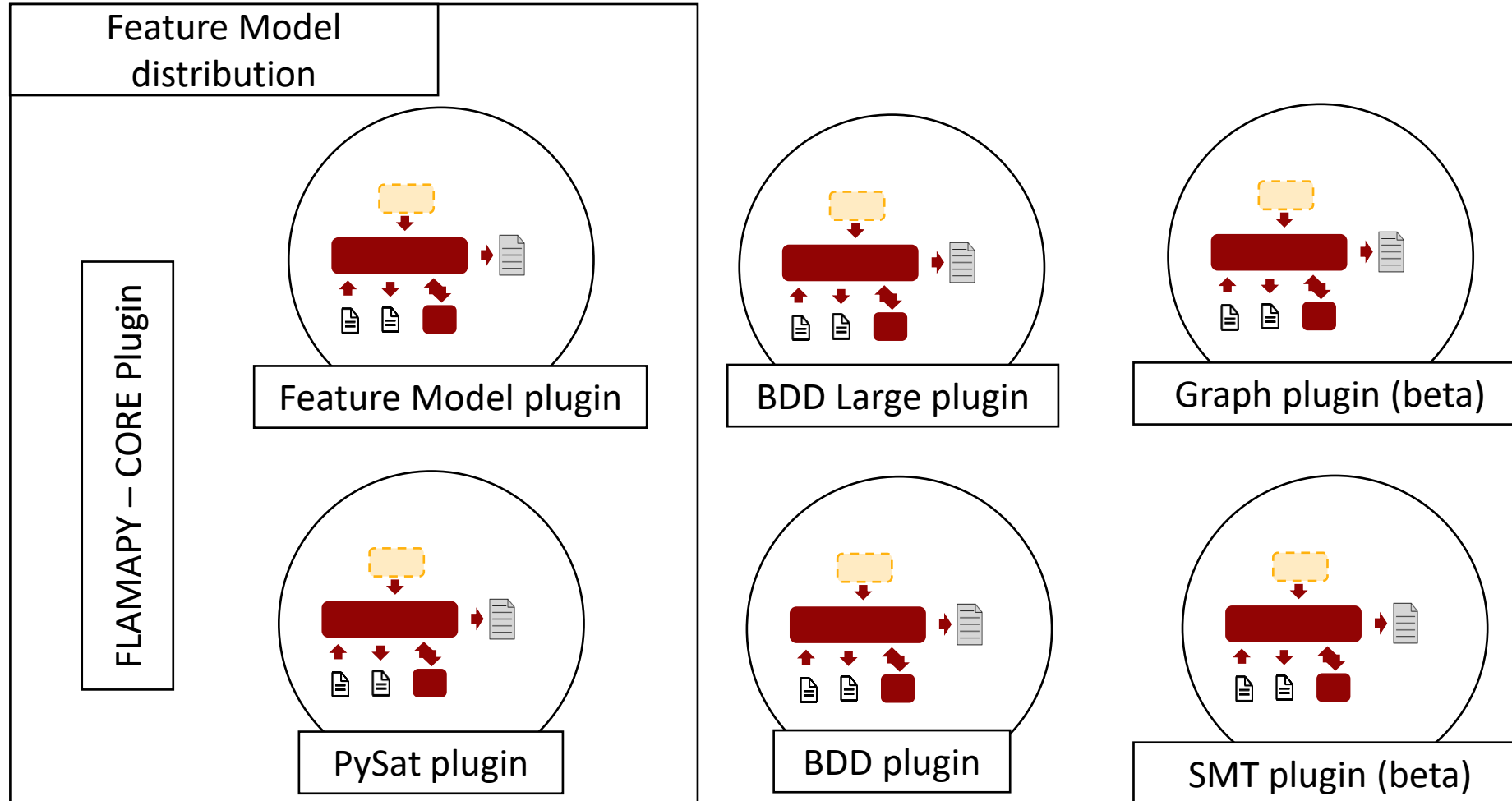
Flama: Resolving dependencies

Pypi to the rescue and the concept of plugins/packages

One plugin → 1--* metamodels



What do we have now?



flamapy



University of Seville

The university of Seville is currently providing support for the core and the pysat plugins.



University of Málaga

The university of Malaga is currently providing support for the core and the bdd plugins.



University of Graz

The university of Graz is currently providing support for the operations regarding the explanation of errors in feature modeling.



Universidad Nacional de Educación a Distancia

The UNED is currently providing support for the implementation of BDD techniques that cope with colossal feature models.

